

What happens if a list is put in a variable?

Copying the value in a variable creates two independent copies of the value. This works if the value is a number, but what about other types of value? If a variable contains a list it works a bit differently.

1 Copy a list

Store the list `[1, 2, 3]` in a variable called `"listA"`. Then store the value of `"listA"` in another variable called `"listB"`. Now both variables contain `[1, 2, 3]`.

```
>>> listA = [1, 2, 3]
>>> listB = listA
>>> print('listA =', listA, 'listB =', listB)
listA = [1, 2, 3] listB = [1, 2, 3]
```

Use square brackets to create a list

This prints out the variable names alongside their values to see what's inside them

`"listA"` and `"listB"` both hold the same value

This changes the second item in the list because lists count from zero

2 Change list A

Change the value in `"listA[1]"` to 1,000. `"listB[1]"` now contains 1,000 as well. Changing the original list has changed the copy of the list too.

```
>>> listA[1] = 1000
>>> print('listA =', listA, 'listB =', listB)
listA = [1, 1000, 3] listB = [1, 1000, 3]
```

The second item of both `"listA"` and `"listB"` has been changed

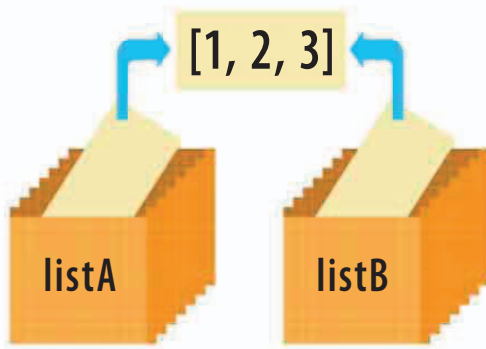
This is the third item in the list

3 Change list B

Change the value of `"listB[2]"` to 75. `"listA[2]"` is now 75 as well. Changing the copy of the list has changed the original list as well.

```
>>> listB[2] = 75
>>> print('listA =', listA, 'listB =', listB)
listA = [1, 1000, 75] listB = [1, 1000, 75]
```

The third item of both `"listA"` and `"listB"` has been changed



△ What's going on?

A variable containing a list doesn't hold the list itself, just a link to it. Copying the value in `"listA"` copies the link. So both `"listA"` and `"listB"` contain a link to the same list.

EXPERT TIPS

Copying lists

To make a separate copy of a list, use the `"copy"` function. `"listC"` will contain a link to a completely new list whose values are copies of those in `"listA"`. Changing `"listC"` won't change `"listA"`, and changing `"listA"` won't change `"listC"`.

```
>>> listC = listA.copy()
```